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Early Prediction of Parkinson's Disease using Voice Biomarkers

Raja Muthiah
Department of CSE
Kalasalingam Academy of Research and Education
Anand Nagar, Krishnankoil-626126, Tamilnadu, India.
mraja@gmail.com

A VENKATA PRANIDEEP
Department of CSE
Kalasalingam Academy of Research and Education
Anand Nagar, Krishnankoil-626126, Tamilnadu, India.
praneethpranideep@gmail.com

B SIVA MANASA REDDY
Department of CSE
Kalasalingam Academy of Research and Education
Anand Nagar, Krishnankoil-626126, Tamilnadu, India
Sivamanasreddy999@gmail.com

OTOORU KULLAYAPA
Department of CSE
Kalasalingam Academy of Research and Education
Anand Nagar, Krishnankoil-626126, Tamilnadu, India.
otoorukullayapa@gmail.com

KALLURI HARISH
Department of CSE
Kalasalingam Academy of Research and Education
Anand Nagar, Krishnankoil-626126, Tamilnadu, India.
harishchowdary33@gmail.com

Abstract Parkinson's disease (PD) is a progressive disorder of the nervous system issues that affect movement and can also affect people's speech. Changes in voice patterns are often visible early on in the condition, making the analysis of speech useful in early detection of the condition. This project focuses on the prediction of PD using machine learning models trained from voice-based biomarkers from UCI Parkinson dataset. The dataset includes a few acoustic features such as jitter shimmer and pitch frequency which helps to capture the vocal instability usually present in PD patients. Before the training the data is standardized, and two classification models are developed - Random Forest and Support Vector Machine (SVM). Hyperparameter Tuning and Cross Validation are used to get better performance. The Random Forest model has shown high accuracy when compared to SVM with around 95% accuracy and recall of around 97% and hence it can be used for early screening. Feature importance results show that jitter and shimmer play a major role in identifying Parkinson's disease using voice samples.

Keywords— Parkinson's Disease, Voice Biomarkers, Machine Learning, Random Forest, Support Vector Machine (SVM), Feature Importance, Early Diagnosis, Speech Analysis, UCI Dataset, Telemedicine.

I. INTRODUCTION

This includes the physical, emotional, social, spiritual, and Financial burden of taking care of the patients with PD. The review describes some of the unique features of Parkinson's, including the following: Motor and non-motor changes often lead to increases in patient needs and dependence on caregivers. Authors synthesis evidence about variables related to patient and caregiver with increased burden, including: symptom severity, time spent caregiving and financial stress. They further discuss negative consequences to the health and quality of the caregivers, such as heightened anxiety and susceptibility to depression. Further in the paper, the interventions to decrease burden to caregivers and update to help clinicians identify and Address this problem in post pandemic era [1]. This Study explores how motor rhythm disturbances are evident in PD patients and hypothesises that dysrhythmia is not uncommon. and can easily be identified using digital techniques. Participants completed the tapping tasks via a special smartphone program that permitted quantitative analysis of the timing of the movement and the coordination of its movement. Authors reported that PD we found prominent rhythmicity deficits in patients compared with healthy controls in supporting the dysrhythmia as a distinctive and quantifiable motor symptoms. An App-based test that can provide a scalable, objective method for screening for and monitoring motor dysfunction in Parkinson's disease [2].

The proposed method is a systematic feature selection framework for selecting the most discriminating features from biomedical data and thus helps in increasing the classification performance by decreasing the computation complexity. It filters remove redundant and irrelevant features at different stages beforehand. Training the model. This framework performs systematic feature selection of important biomedical features in order to improve classification performance while reducing computation; multi-stage filtering is used to eliminate redundancies in the pre-training process. Tested on various data/classifiers, it outperforms baselines in a robust manner on small data, making it appropriate for clinical settings for objective and scalable PD early detection - thus providing smart selection role in diagnostics [3].

This project basically consist of whole end to end machine learning pipeline for the early detection of Parkinson's disease based on voice biomarkers presented on UCI wages dataset. The workflow starts with exploratory data analysis which deals with exploring feature pattern, class imbalance and correlation among variables. Secondly, the pre
executed such as normalization, missing values, and

elimination of redundant features to get reliable and consistent information. The dataset is then divided into train and test using stratified sampling in which both the train set and test dataset have balanced class ratio. Two classification models are trained using the prepared data named as Random Forest and Support Vector Machine. Their hyperparameters are tuned using grid search with cross validation recall is a key considerate their in order to detect as cases of Parkinson's as possible. The model performance is measured in terms of visual and statistical measurements (of confusion matrices, ROC curves and feature importance plots). The results of the analysis identify jitter and shimmer as important predictors which contribute highly to classification. In order to further ensure that the pipeline is in fact usable in the real world, it is further evaluated on unseen samples, which results in not only class predictions, but also confidence scores. The last trained model with the preprocessing is saved for further use in the deployment of digital health systems and telemedicine systems.[4]

II. LITERATURE SURVEY

[5] It should be Evolved into the widely accepted eponym *Parkinson's disease*, tracing its historical adoption and reflecting on the broader role of eponyms in medical naming.

[6] In societal/contextual issues (including the COVID- 19 pandemic, stigma and limited support) — and that although some interventions show promise, there remains a need for more rigorous, targeted strategies to reduce this burden

[7] Developed a wearable (wrist worn) system that records motion activities and computes two digital biomarkers -- motion and a rest-tremor index -- that can help differentiate the early Parkinson's disease patients from healthy controls.

[8] It should be detected using a smartphone-based tapping test. PD patients had greater tap-to-tap variability than healthy controls, and tapping speed slightly declined over one year, suggesting potential for digital motor monitoring in PD.

[9] It will study presents a novel multi-level feature selection framework applied to voice-based recordings that, when combined with an optimized KNN classifier, achieves high accuracy (≈95.4%) in detecting Parkinson's disease.

[10] A hybrid approach has been devised which integrates Optimized Learning Vector Quantization (OLVQ1) and Adaptive Neuro-Fuzzy Inference System (ANFIS). This approach utilizes the information about vocal features and it is capable of predicting both of the scores of Motor UPDRS and Total UPDRS with high degree of accuracy (highest R^2values ~0.99 and lowest RMSE ~0.56.

[11] A multitask deep neural network was developed based on the features extracted from the voice samples. The model combines the two tasks of classifying if their Parkinson's is severe or non-severe and predicting the motor and total UPDRS scores. The approach was able to exhibit very high accuracy in both classification and regression tasks.

[12] Parkinson's disease imposes a significant and increasing economic burden worldwide, with total costs rising sharply as disease severity progresses—mainly due to hospitalization, medication, and productivity losses—highlighting the urgent need for better management strategies and healthcare planning.

[13] This study presents a multi-cohort, cross-sectional design of investigation with a particular focus in gathering and analysis for digital biomarkers, abnormalities, associated with cognitive and behavioral patterns in patients with neurodegenerative disorders. The findings are expected to aid in the earlier diagnosis and better monitoring of the disease and disease progression.

[14] Evidence that machine learning has the ability to differentiate between patients motor status with impeccable accuracy is provided when the NNC methodology was applied to a digital biomarker dataset related to Parkinson's diseases to showcase the potential of digital biomarkers to monitor the disease objectively.

[15] This study confirms that smartphone-based motor testing is able to distinguish in a clear way between the group of people with idiopathic REM sleep behavior disorder, healthy participants, and people diagnosed with Parkinson's disease. This shows its potential use as a practical tool for the early detection and continuous monitoring of motor impairment of neurodegenerative conditions.

[16] Real-life gait performance measured by wearable sensors are able to be an reliable digital biomarker to detect and monitor motor fluctuations of patients with PD in an out-of-clinic environment.

[17] Digital biomarkers will be collected using smartphone-based remote assessment in order to measure both motor and non-motor symptoms of individuals with PD in a reliable way, thus enabling large-scale and accessible monitoring.

[18] Digital biomarkers obtained using remote evaluation via smartphones can reliably quantify the motor and non-motor symptoms in individuals with Parkinson's disease; making large-scale and accessible monitoring possible.

[19] Findings from the m-Power dataset reveal that digital biomarkers from the dataset are strongly correlated (test-retest), and are stable over time. This underpins their possible use in monitoring the progression of Parkinson's disease in the open world.

[20] Research shows that non-motor symptoms of Parkinson's disease—including cognitive, autonomic, and psychiatric changes—are linked to distinct neuropathological mechanisms. This indicates that Parkinson's disease affects multiple brain regions and is not limited to dysfunction of the dopaminergic system.

[21] That voice signal analysis can be used to predict and estimate the severity of the disease in Parkinson's, and this provides a noninvasive tool in monitoring disease progression in Parkinson's.

III. METHODOLOGY

The Proposed System Here's a methodology section of a project that you would write on the detection of Parkinson's Disease using biomarkers in voice-in sentence form. Figure 1: Overall Architecture will be here of methodology

A. Voice Data Collection

It receives voice recordings from PD patients and healthy people. The recordings of the speech consist of sustained speech of vowels such as /a/, /e/, /i/, and short sentences of speech to record different speech features.

B. Preprocessing

Get rid of background noise and artifacts in recordings; scale audio signals to a conventional amplitude range, and use some trick of smoothing or filter to enhance the sound.

C. Feature Extraction

Extract acoustic features that reflect vocal changes, such as:

- Jitter
- Shimmer
- Harmonic-to-Noise Ratio
- Mel-Frequency Cepstral Coefficients

D. Feature Selection

Analyze extracted features to identify the most significant ones for classification, using either statistical methods or model-based selection; for example, random forest importance, L1-norm SVM. Reduce the dimensions to improve accuracy and reduce computation.

E. Model Traninig And Machine Learning

Segmentation of the dataset into the training dataset and the test dataset, which usually means 80:20, demands stratification for balanced classes. Train the machine learning classifiers Random forest and SVM. Use Grid Search with cross- validation in order to optimize the hyperparameters for better performance.

F. Model Evalution

Evaluate these models with regards to accuracy, precision, recall, F1- score and ROC- AUC. View results- Areas of confusion matrice and ROC.

G. Prediction on New Data

PD detection on new voice samples by means of the trained PD model. Get the predictions as well as the probability/confidence scores.

H. Model Saving and Deployment

Save the trained model and scaler to be used in the future in a web or mobile application. This enables the detection of PD in a non- invasive and early way in reality

I. Summary of Methodology

These are neatly divided into six important stages which are simply explained to become perfectly in Table 1 in order to show the detail manner.

STAGE	DESCRIPTION
Voice Input	In this stage, voice samples are recorded from both Parkinson's victims and healthy people.
Preprocessing	The recorded voice signals are cleaned to remove background noise and distortions.
Feature Extraction	These features lock in the irregularities of the vocal folds that are common signs of Parkinson's Disease.
Feature Selection	The extracted features are analyzed to identify the most relevant ones that contribute significantly to the prediction.
Machine Learning Classification	Machine learning algorithm such as Random Forest and Support Vector Machine (SVM) are trained by chosen features.
Result and Evaluation	These performance metrics such as accuracy, precision, recall, F1-score.

Table I: Methodology of the Proposed Parkinsons Diseases

IV. IMPLEMENTATION

This project was done with the assistance of Python programming language using multiple machine learning libraries such as Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib, etc. The process begins with the loading of the Parkinson's data set in UCI Machine Learning Repository that has several of the acoustic features derived from voice recordings of healthy people and patients with Parkinson's. The dataset is explored. to get idea about its structure, missing values in it, o get idea about feature correlation in it. Then, some pre-processing was done by eliminating the columns that were not needed like "name" & ensuring that the missing values were taken care of & features were normalized using the Standard Scaler to make the ranges of the features the same. The processed data is then divided in 80: 20 ratio of training data and testing data set, respectively using stratified sampling in order to maintain the class balance.

The trained models are then used for evaluation on the test dataset based on the different performance metrics: accuracy, precision, recall, F1-score and ROC-AUC. Visualizations such as confusion matrix and ROC curves are produced which helps in determining the efficiency of each the models. The highest performance gained using the Random Forest making it best model for early detection.

Finally, the best model and scaler are saved using Job lib for future deployment, allowing the system to predict Parkinson's Disease for new voice inputs. This implementation provides a complete and practical pipeline for early and non-invasive diagnosis using voice biomarkers in (Fig 1 it shows the total implement

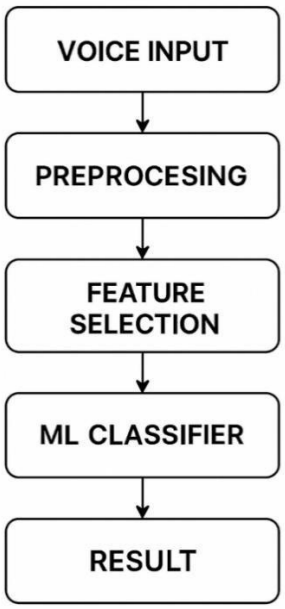

Fig 1: Flow chart of implementation

V. RESULT AND DISCUSSION

The proposed system was able to successfully work with python language and tested with the dataset of Parkinson's which was downloaded from the UCI Machine Learning Repository. Two machine learning algorithms - Random Forest and Support Vector Machine (SVM) were trained and test to identify individuals as being healthy or suffering from Parkinson's based on the voice biomarker extracted

It is evident that the dataset contains a higher number of Parkinson's cases compared to healthy cases. Specifically, there are around 50 healthy samples and approximately 150 Parkinson's samples. This indicates an imbalanced dataset, which is an important consideration for training machine learning models to avoid biased predictions (Fig 2: it shows the healthy and disease while be shown).

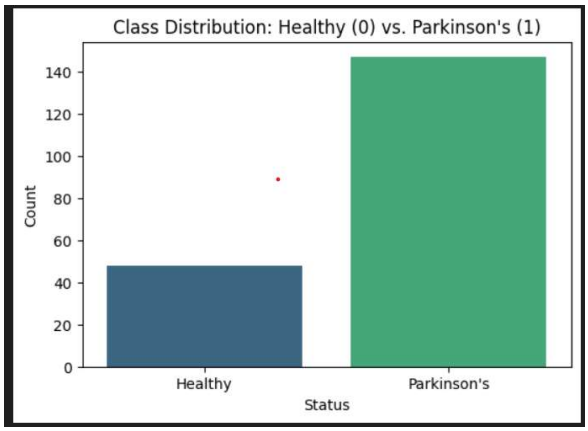

Fig 2: Status Of Healthy Persons And Parkinsons Disease Person

The confusion matrix helps us to understand the performance of the Random Forest model in classifying the individuals to be eitherHealthy or having PD or Parkinson's Disease (PD)

While this model made the right diagnosis on 8 cases of Healthy people and 28 cases of PD, it misdiagnosed 2 cases of Healthy people has PD and one case of PD patients as Healthy people. Overall, the model is very accurate for the cases of PD with high accuracy in the correct detection of the cases. Fig 3 it will show the how the confusion matrix of random forest will look like

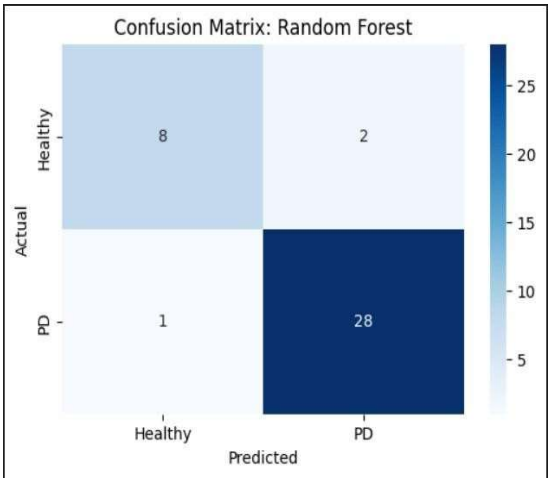

Fig 3: Confusion Matrix of Random Forest

The ROC curve is the performance of the Random Forest model to separate the case of Healthy case and PD case. The curve is high true positive with low false positive strong classification ability. An AUC value of 0.96 indicates that the performance of the model is excellent in differentiating PD patients and Healthy people with great accuracy. Fig 4 will be showing to form of random forest curve.

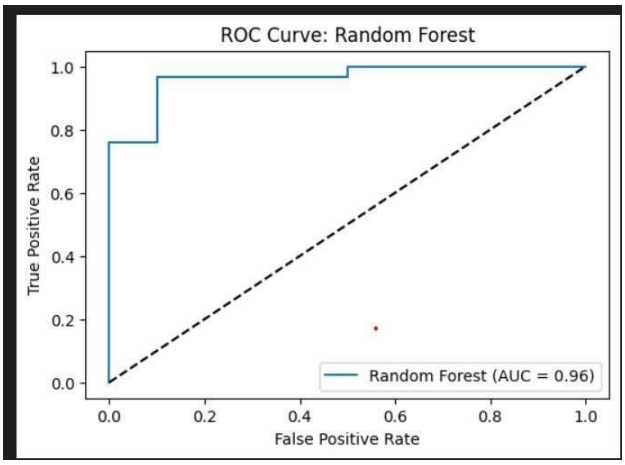

Fig 4: Curve Form

Confusion matrix is used to describe the performance of SVM model in terms of the classification of the subjects as Healthy or has PD. It correctly diagnosed 29 cases of PD however all the 10 cases of Healthy people were misdiagnosed with PD, which means that even though we have a SVM model with a high sensitivity of finding PD, our model fails to be specific about identifying Healthy cases. Fig 5 will be shows confusion matrix of SVM

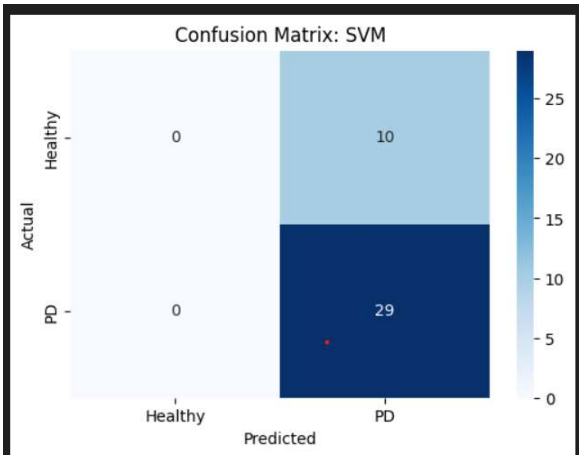

Fig 5: Support Vector Machine

The ROC curve is used to explain the work of the Support Vector Machine model to classify the Healthy and PD cases. The curve demonstrates that the actual positive rate is high throughout most of the false positive rates and thus a measure of a good ability of discriminating. The AUC value of 0.92 indicates that the general performance of SVM model is good that can efficiently distinguish between PD and Healthy Indivi. with lesser strength as compared to Random Forest model. Fig 6 will visualize in curve form of SVM.

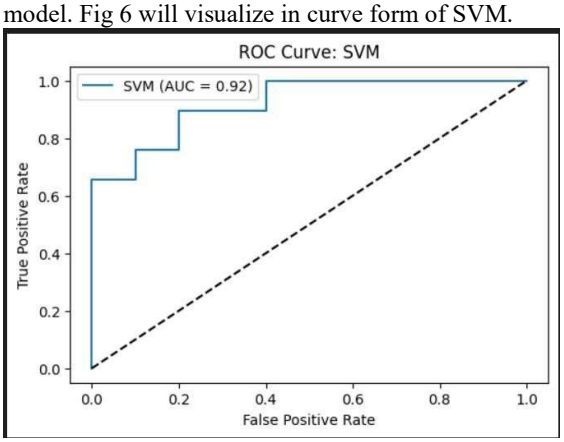

Fig 6: Curve Form

ROC curves of Random Forest and SVM model are given below showing good comparison between performance of Random forest and SVM model in discriminating between the healthy and PD patients. It is seen that Random Forest model is playing a better role in terms of classification capabilities with the AUC value which is 0.96 to correctly identify the individuals who has PD and the false positives remains minimal as compared to SVM model. In this regard, the latter with an AUC of 0.92 also presents an excellent performance with a slightly lower discriminative power. A more steeper of a curve the Random Forest has a higher rate of true positives at almost every level of false positive rate. While both the models give good to excellent performances, if we compare both Random forest model has high predictive power and reliability in PD detection. (Fig 7 will be shows in comparison table of Random Forest and Support vector Machine.

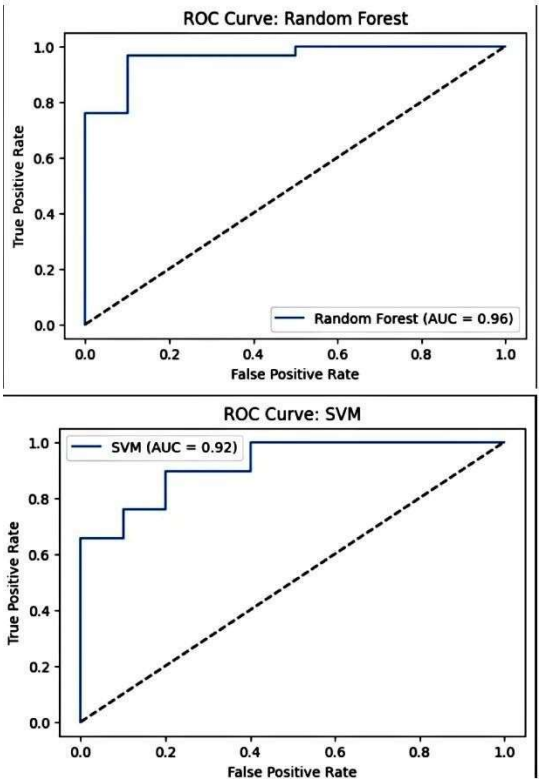

Fig 7: Comparison Table of Support Vector Machine And Random Forest

Fig. below is the top 10 Most informative voice biomarkers using Random Forest model determined to predict PD. Among them, the highest elements of importance are those for PPE and spread1; they have the most contribution to the decision of the model. Other important features, also in the set, are MDVP Fo(Hz), NHR and Jitter:DDP which suggests that variation in pitch and voice frequency is the key in distinguishing the PD patients from Healthy individuals. (Fig 8 will be shows in taking 10 biomarkers in data.

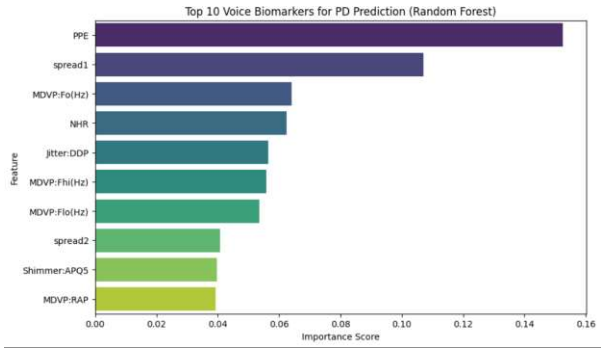

Fig 8: Voice Biomarkers of prediction

VI. CONCLUSION AND FUTURE SCOPE

The presented work, in this publication presents the entire pipeline for the early detection of PD using voice recording. In particular, classical machine learning models especially Random forest and SVM are very good with a high accuracy and recall for the study therefore making them viable for screening purposes. Feature analysis has shown that important voice biomarkers such as jitter, shimmer and HNR are important in the detection of vocal irregularities

The pipeline implemented including preprocessing,

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preprocessing, Feature extraction and feature selection, model training. evaluation and pipeline implemented offers is a reproducible and is interpretive approach.

In future such deep learning techniques like CNN, RNN or Transformer based model may be studied in order to consider to automatically learn some complex features from the voice data to increase the prediction accuracy. Larger and more diverse data with multi-lingual and real-world data recorded will result in models being more generalized. Integration with mobile and IoT to real-time monitoring may possibly lead to early detection of PD using smart phones or wearable devices. Longitudinal studies can also be helpful to determine the path of a disease and the efficiency of i.e. early intervention.

VII. **REFERENCES**

[1] M. Weatherall, 'Parkinson's eponym', Lancet Neurol. vol. 23, no. 2, p. 143, Feb. 2024, doi: 10.1016/S1474-4422(23)00323-X.

[2] W. W. Aamodt et al., 'Caregiver Burden in Parkinson Disease: A Scoping Review of the Literature from 2017-2022', J. Geriatr. Psychiatry Neurol., vol. 37, no. 2, pp. 96–113, Mar. 2024, doi: 10.1177/08919887231195219.

[3] Battista L, Romaniello A. A new wrist-worn tool supporting the diagnosis of parkinsonian motor syndromes. Sensors 2024;24. https://doi.org/10.3390/ s24061965.

[4] Marsili L, Abanto J, Mahajan A, Duque KR, Chinchihualpa Paredes NO, Deraz HA, et al. Dysrhythmia as a prominent feature of Parkinson's disease: an app-based tapping test. J Neurol Sci 2024;463. https://doi.org/10.1016/j.jns.2024.123144.

[5] F. Demir, K. Siddique, M. Alswaitti, K. Demir, and A. Sengur, 'A Simple and Effective Approach Based on a Multi-Level Feature Selection for Automated Parkinson's Disease Detection', J. Pers. Med., vol. 12, no. 1, Art. no. 1, Jan. 2022, doi: 10.3390/jpm12010055.

[6] W. A. Zogaan et al., 'A combined method of optimized learning vector quantization and neuro-fuzzy techniques for predicting uni ed Parkinson's disease rating scale using vocal features', MethodsX, vol. 12, p. 102553, Jun. 2024, doi: 10.1016/j.mex.2024.102553.

[7] M. T. García-Ordás, J. A. Benítez-Andrades, J. Aveleira-Mata, J.-M. Alija-Pérez, and C. Benavides, 'Determining the severity of Parkinson's disease in patients using a multi task neural network', Multimed. Tools Appl., vol. 83, no. 2, pp. 6077–6092, Jan. 2024, doi: 10.1007/s11042-023-14932-x.

[8] Chaudhuri KR, Azulay J-P, Odin P, Lindvall S, Domingos J, Alobaidi A, et al. Economic burden of Parkinson's Disease: a multinational, real-world, cost-of- illness study. Drugs Real World Outcomes 2024. https://doi.org/10.1007/s40801- 023-00410-1.

[9] Crook-Rumsey M, Daniels SJC, Abulikemu S, Lai H, Rapeaux A, Hadjipanayi C, et al. Multicohort cross-sectional study of cognitive and behavioural digital biomarkers in neurodegeneration: the Living Lab Study protocol. BMJ Open 2023; 13:e072094. https://doi.org/10.1136/bmjopen-2023-072094.

[10] Tsoulos IG, Mitsi G, Stavrakoudis A, Papapetropoulos S. Application of machine learning in a parkinson's disease digital biomarker dataset using Neural Network Construction (NNC) methodology discriminates patient motor status. Front ICT 2019;6. https://doi.org/10.3389/fict.2019.00010.

[11] Arora S, Baig F, Lo C, Barber T, Lawton M, Zhan A, et al. Smartphone motor testing to distinguish idiopathic REM sleep behavior disorder, controls, and PD. NEUROLOGY 2018;91:1528https://doi.org/10.1212/WNL.0000000000006366.

[12] Evers LJW, Raykov YP, Krijthe JH, Silva De Lima AL, Badawy R, Claes K, et al. Real-life gait performance as a digital biomarker for motor fluctuations: the Parkinson@Home validation study. J Med Internet Res 2020;22. https://doi.org/ 10.2196/19068.

[13] Goni M, Eickhoff SB, Far MS, Patil KR, Dukart J. Smartphone-based digital biomarkers for Parkinson's disease in a remotely-administered setting. IEEE Access 2022;10:28361–84. https://doi.org/10.1109/ACCESS.2022.3156659

[14] Johnson S, Kantartjis M, Severson J, Dorsey R, Adams J, Kangarloo T, et al. Wearable sensor-based assessments for remotely screening early-stage Parkinson's disease. SENSORS 2024;24. https://doi.org/10.3390/s24175637.

[15] Far MS, Eickhoff SB, Goni M, Dukart J. Exploring test-retest reliability and longitudinal stability of digital biomarkers for parkinson disease in the m-power data set: cohort study. J Med Internet Res 2021;23. https://doi.org/10.2196/ 26608.

[16] Dickson DW, Fujishiro H, Orr C, DelleDonne A, Josephs KA, Frigerio R, et al. Neuropathology of non-motor features of Parkinson disease. Parkinsonism Relat Disord 2009;15(Suppl 3):S1–5. https://doi.org/10.1016/S1353-8020(09)70769-2.

[17] Hemmerling, D., & Wojcik-Pedziwiatr, M. (2022), Prediction and estimation of Parkinson's disease severity based on voice signal, Journal of Voice, 36(3), 439-e9.